Register spill study: test vs. train vs. ref (SimPoint)

L1D 4 KiB, L1I 32 KiB, L2 32 KiB, 64 B lines, LRU · metric schema 9 · gem5 TimingSimpleCPU · 4 November 2026

Each value pools all completed runs of the benchmark for that input: sum of numerators / sum of denominators. *runs t/tr/r* = completed runs in test / train / ref. ref values are SimPoint estimates (up to 30 points of 100 M instructions after a 500 M warmup); test and train are full runs. *pooled* = all completed runs of the input together (large runs dominate); *mean* = average of the benchmark values over the benchmarks that have a value in all three inputs, each benchmark weighted equally. A benchmark's runs differ between inputs (e.g. perlbench test = makerand, train = 5 scripts, ref = 3 scripts). Formulas: last page.

Program, compiler spill (does not depend on the cache)

		spill/ROI			spill ld/load			spill st/store		
benchmark	runs t/tr/r	test	train	ref	test	train	ref	test	train	ref
perlbench	1/5/3	0.08%	2.53%	5.33%	0.00%	4.68%	6.68%	0.50%	7.77%	19.51%
gcc	1/3/5	0.84%	1.21%	1.01%	2.18%	2.77%	2.27%	3.58%	4.53%	2.88%
bwaves	2/2/4	3.55%	3.01%	2.55%	9.47%	7.73%	6.31%	18.87%	18.85%	18.88%
mcf	1/1/1	1.58%	1.61%	1.40%	2.11%	2.27%	2.06%	16.47%	13.28%	7.74%
cactuBSSN	1/1/1	21.32%	21.33%	20.65%	51.31%	51.32%	47.95%	55.47%	55.49%	56.30%
namd	1/1/1	3.13%	3.22%	3.27%	7.91%	8.11%	8.21%	17.27%	17.74%	17.87%
parest	1/1/1	0.22%	0.18%	0.16%	0.56%	0.45%	0.43%	1.65%	1.28%	0.62%
povray	1/1/1	0.27%	0.27%	0.27%	0.45%	0.46%	0.46%	0.93%	0.95%	0.95%
lbm	1/1/1	9.67%	9.69%	10.48%	27.79%	27.97%	32.28%	35.34%	35.99%	44.97%
omnetpp	1/1/1	0.09%	0.01%	0.00%	0.19%	0.02%	0.00%	0.36%	0.03%	0.00%
wrf	1/1/1	0.91%	0.92%	0.91%	3.41%	3.44%	3.38%	2.49%	2.61%	2.59%
xalancbmk	1/1/1	0.23%	0.44%	0.13%	0.53%	1.00%	0.26%	0.92%	1.04%	0.80%
x264	1/1/3	6.24%	6.30%	6.42%	13.47%	13.42%	13.69%	39.69%	40.78%	40.32%
blender	1/1/1	1.16%	0.31%	0.38%	2.72%	0.55%	0.88%	4.44%	4.03%	3.23%
cam4	1/1/1	2.11%	2.49%	3.14%	7.90%	9.15%	11.02%	7.04%	7.96%	11.03%
deepsjeng	1/1/1	2.78%	2.99%	2.92%	9.33%	9.41%	9.17%	15.84%	15.98%	16.25%
imagick	1/1/1	0.02%	0.04%	0.23%	0.08%	0.21%	0.12%	0.05%	4.19%	3.44%
leela	1/1/1	0.36%	0.43%	0.38%	1.25%	1.55%	1.36%	1.66%	2.04%	1.80%
nab	1/2/1	7.90%	7.33%	5.20%	26.29%	24.99%	15.37%	39.86%	35.42%	34.02%
exchange2	1/1/1	5.03%	4.84%	5.23%	10.75%	10.67%	10.55%	12.58%	12.41%	12.68%
fotonik3d	1/1/1	3.84%	3.99%	3.36%	10.81%	10.66%	9.32%	9.25%	12.65%	9.25%
roms	1/0/1	4.79%	—	4.37%	15.22%	—	13.83%	9.67%	—	8.97%
xz	12/2/3	1.99%	2.36%	2.22%	7.05%	8.36%	8.26%	14.21%	14.50%	24.23%
pooled		5.17%	3.60%	3.78%	12.33%	9.44%	9.06%	27.44%	17.30%	17.53%
mean (22 benchmarks)		3.33%	3.43%	3.44%	8.89%	9.05%	8.64%	13.57%	14.07%	14.97%

Program, spill + csave (does not depend on the cache)

		(spill+csave)/ROI			(spill+csave) ld/load			(spill+csave) st/store		
benchmark	runs t/tr/r	test	train	ref	test	train	ref	test	train	ref
perlbench	1/5/3	18.63%	17.68%	21.33%	28.89%	31.56%	34.71%	59.00%	56.14%	65.16%
gcc	1/3/5	12.72%	16.27%	14.46%	34.38%	34.50%	30.12%	52.28%	66.41%	44.45%
bwaves	2/2/4	3.55%	3.01%	2.55%	9.47%	7.73%	6.31%	18.87%	18.85%	18.88%
mcf	1/1/1	2.74%	2.75%	2.33%	3.87%	4.04%	3.60%	27.40%	21.88%	12.43%
cactuBSSN	1/1/1	21.42%	21.43%	20.67%	51.49%	51.50%	48.00%	55.80%	55.81%	56.39%
namd	1/1/1	3.14%	3.23%	3.27%	7.94%	8.13%	8.22%	17.37%	17.81%	17.90%
parest	1/1/1	0.81%	0.49%	0.64%	1.48%	0.93%	1.19%	12.55%	7.38%	7.86%
povray	1/1/1	14.17%	14.26%	14.26%	22.21%	22.21%	22.18%	55.05%	55.40%	55.24%
lbm	1/1/1	9.67%	9.69%	10.48%	27.79%	27.97%	32.28%	35.34%	35.99%	44.97%
omnetpp	1/1/1	12.82%	12.59%	12.75%	28.51%	24.64%	24.67%	53.40%	48.38%	48.13%
wrf	1/1/1	1.59%	1.59%	1.52%	4.94%	4.97%	4.76%	7.83%	8.09%	7.57%
xalancbmk	1/1/1	8.96%	18.29%	4.92%	16.83%	32.42%	8.32%	45.24%	63.05%	39.84%
x264	1/1/3	8.48%	8.75%	8.89%	18.06%	18.41%	18.75%	54.69%	57.31%	56.61%
blender	1/1/1	9.37%	3.18%	3.72%	20.20%	5.93%	6.96%	39.50%	39.04%	42.51%
cam4	1/1/1	4.52%	4.59%	4.99%	14.20%	14.65%	15.77%	21.40%	19.40%	21.23%
deepsjeng	1/1/1	7.31%	8.19%	8.11%	22.43%	23.80%	23.58%	46.66%	48.29%	49.72%
imagick	1/1/1	4.16%	0.25%	5.33%	11.90%	0.82%	12.37%	27.17%	48.78%	46.11%
leela	1/1/1	6.11%	6.08%	6.07%	15.91%	15.99%	15.90%	44.30%	46.72%	45.53%
nab	1/2/1	8.63%	8.25%	6.75%	27.88%	27.01%	18.74%	47.43%	45.59%	49.89%
exchange2	1/1/1	6.23%	6.03%	6.08%	13.14%	13.09%	12.17%	15.81%	15.72%	14.84%
fotonik3d	1/1/1	3.87%	4.00%	3.38%	10.86%	10.69%	9.37%	9.40%	12.75%	9.40%
roms	1/0/1	4.88%	—	4.44%	15.37%	—	13.96%	10.29%	—	9.48%
xz	12/2/3	3.68%	4.71%	3.21%	11.80%	15.14%	11.17%	30.33%	33.39%	39.06%
pooled		7.17%	7.34%	7.49%	16.47%	17.15%	16.44%	39.99%	41.91%	39.26%
mean (22 benchmarks)		7.84%	7.97%	7.53%	18.37%	18.01%	16.78%	35.31%	37.37%	36.08%

Miss rates: small-2way (L1D 32 sets x 2, L2 64 sets x 8)

		L1D			spill ld L1			spill st L1			L2		
benchmark	runs t/tr/r	test	train	ref	test	train	ref	test	train	ref	test	train	ref
perlbench	1/5/3	5.72%	6.72%	6.44%	0.00%	3.23%	1.05%	2.72%	3.57%	2.69%	0.00%	27.70%	16.94%
gcc	1/3/5	7.17%	9.83%	10.70%	0.67%	2.42%	2.73%	2.17%	4.91%	5.12%	32.52%	36.68%	49.05%
bwaves	2/2/4	41.07%	42.18%	42.96%	2.70%	3.24%	4.06%	0.33%	0.27%	0.19%	31.22%	27.26%	24.23%
mcf	1/1/1	25.50%	23.82%	22.05%	5.57%	7.04%	7.93%	8.28%	8.49%	7.61%	75.88%	75.08%	72.44%
cactuBSSN	1/1/1	15.76%	15.77%	15.64%	4.45%	4.46%	4.39%	3.91%	3.90%	3.60%	64.32%	64.32%	64.77%
namd	1/1/1	12.89%	12.79%	12.51%	5.49%	5.33%	3.31%	1.31%	1.39%	1.36%	25.71%	25.42%	25.47%
parest	1/1/1	11.59%	11.51%	11.55%	9.58%	4.71%	3.60%	3.51%	2.95%	3.31%	57.67%	61.90%	63.65%
povray	1/1/1	10.14%	10.10%	10.32%	7.41%	7.63%	7.39%	20.68%	20.37%	19.42%	41.21%	41.87%	40.97%
lbm	1/1/1	23.74%	23.36%	19.64%	0.42%	0.42%	0.44%	2.26%	2.11%	1.75%	88.75%	88.88%	86.89%
omnetpp	1/1/1	12.56%	16.55%	16.65%	0.25%	0.25%	0.09%	0.86%	0.87%	2.02%	26.59%	28.83%	27.55%
wrf	1/1/1	18.06%	17.59%	17.88%	9.40%	9.17%	9.00%	6.61%	6.10%	5.95%	22.28%	22.33%	21.73%
xalancbmk	1/1/1	13.66%	12.40%	17.45%	10.00%	20.48%	10.52%	18.70%	29.68%	15.82%	33.74%	12.36%	68.78%
x264	1/1/3	2.33%	2.33%	2.35%	0.39%	0.39%	0.33%	0.95%	0.94%	0.92%	26.97%	28.09%	28.97%
blender	1/1/1	5.34%	7.42%	8.22%	2.28%	3.84%	2.76%	3.16%	7.01%	2.06%	19.43%	11.64%	17.84%
cam4	1/1/1	10.24%	11.20%	11.80%	2.66%	2.59%	3.18%	2.91%	2.62%	3.08%	52.99%	54.29%	50.53%
deepsjeng	1/1/1	9.98%	9.80%	9.86%	2.86%	2.94%	2.97%	7.61%	7.87%	7.98%	6.89%	6.58%	5.80%
imagick	1/1/1	4.42%	5.02%	2.17%	10.07%	34.99%	3.94%	69.57%	0.47%	0.03%	81.36%	94.86%	96.75%
leela	1/1/1	6.90%	6.41%	6.48%	9.70%	7.02%	4.29%	0.42%	0.20%	0.04%	20.72%	16.65%	18.20%
nab	1/2/1	6.91%	6.96%	7.99%	1.39%	1.13%	1.57%	2.12%	1.32%	2.82%	4.63%	37.99%	40.00%
exchange2	1/1/1	2.50%	2.26%	1.83%	2.74%	2.05%	1.62%	3.56%	2.70%	2.78%	1.08%	1.44%	0.85%
fotonik3d	1/1/1	17.86%	17.95%	16.67%	5.71%	5.70%	6.14%	0.06%	0.06%	0.17%	41.35%	38.50%	41.15%
roms	1/0/1	21.00%	—	21.55%	4.44%	—	4.01%	1.56%	—	1.25%	38.82%	—	38.82%
xz	12/2/3	7.11%	9.38%	10.41%	1.82%	3.30%	3.54%	1.53%	3.67%	1.48%	24.30%	24.22%	32.02%
pooled		8.77%	11.76%	12.40%	2.14%	3.44%	3.33%	1.96%	3.36%	2.81%	38.69%	36.78%	41.46%
mean (22 benchmarks)		12.34%	12.79%	12.80%	4.34%	6.02%	3.86%	7.42%	5.07%	4.10%	35.44%	37.59%	40.66%

Miss rates: small-fa (L1D 1 sets x 64, L2 1 sets x 512)

		L1D			spill ld L1			spill st L1			L2		
benchmark	runs t/tr/r	test	train	ref	test	train	ref	test	train	ref	test	train	ref
perlbench	1/5/3	0.08%	3.88%	4.32%	0.00%	0.17%	0.09%	0.26%	2.24%	2.65%	0.08%	37.77%	22.57%
gcc	1/3/5	5.69%	7.79%	9.08%	0.22%	0.89%	1.10%	1.34%	3.97%	4.25%	38.30%	43.13%	55.99%
bwaves	2/2/4	7.31%	7.29%	7.29%	0.86%	0.74%	0.71%	0.00%	0.00%	0.00%	95.60%	97.27%	97.90%
mcf	1/1/1	24.52%	22.96%	21.77%	3.11%	3.67%	2.59%	6.86%	7.07%	6.13%	78.48%	77.43%	72.93%
cactuBSSN	1/1/1	15.22%	15.23%	15.37%	4.85%	4.85%	4.99%	5.15%	5.15%	4.97%	59.94%	59.95%	61.01%
namd	1/1/1	11.61%	11.49%	11.54%	2.56%	2.42%	2.41%	1.18%	1.26%	1.29%	28.07%	27.80%	27.21%
parest	1/1/1	9.72%	9.87%	9.78%	1.35%	1.06%	1.18%	1.45%	1.25%	1.28%	68.60%	72.07%	75.05%
povray	1/1/1	8.70%	8.60%	8.49%	7.06%	7.07%	7.25%	19.52%	19.32%	18.47%	50.57%	51.48%	50.78%
lbm	1/1/1	21.23%	20.86%	17.10%	0.00%	0.00%	0.00%	0.55%	0.37%	0.02%	99.28%	99.62%	99.98%
omnetpp	1/1/1	11.53%	16.03%	16.24%	0.05%	0.05%	0.08%	0.77%	0.79%	1.36%	27.53%	28.39%	27.41%
wrf	1/1/1	5.88%	5.78%	5.68%	5.43%	5.29%	5.10%	4.49%	4.18%	3.73%	61.76%	61.40%	62.89%
xalancbmk	1/1/1	13.00%	11.44%	16.81%	10.87%	6.74%	11.16%	21.93%	32.17%	22.22%	34.56%	6.67%	70.85%
x264	1/1/3	1.84%	1.80%	1.88%	0.16%	0.15%	0.12%	0.85%	0.83%	0.80%	31.44%	33.73%	32.95%
blender	1/1/1	5.13%	6.70%	8.01%	2.12%	4.77%	2.59%	3.87%	2.35%	1.35%	19.66%	11.68%	17.55%
cam4	1/1/1	8.80%	9.81%	10.39%	1.46%	1.48%	1.80%	2.47%	1.85%	2.11%	59.48%	60.40%	56.58%
deepsjeng	1/1/1	8.08%	7.79%	7.89%	1.43%	1.45%	1.59%	7.23%	7.51%	7.48%	6.11%	6.05%	5.04%
imagick	1/1/1	4.16%	5.01%	2.22%	7.20%	34.95%	3.94%	58.93%	0.40%	0.01%	87.38%	94.98%	100.00%
leela	1/1/1	5.07%	4.49%	4.02%	0.33%	0.20%	0.06%	0.42%	0.20%	0.00%	27.89%	22.42%	28.13%
nab	1/2/1	3.89%	3.91%	3.91%	0.13%	0.06%	0.08%	0.11%	0.21%	0.25%	8.31%	68.09%	82.17%
exchange2	1/1/1	1.54%	1.32%	1.13%	1.69%	1.29%	1.29%	2.61%	1.88%	1.65%	1.79%	2.51%	1.41%
fotonik3d	1/1/1	8.89%	8.39%	8.23%	1.99%	1.99%	2.06%	0.01%	0.01%	0.01%	82.63%	82.20%	83.12%
roms	1/0/1	10.76%	—	10.86%	2.17%	—	1.55%	0.97%	—	0.72%	64.31%	—	67.88%
xz	12/2/3	5.93%	7.81%	9.57%	0.35%	0.70%	0.93%	0.35%	0.54%	0.55%	28.47%	28.62%	34.39%
pooled		5.59%	8.18%	8.19%	1.59%	2.52%	2.52%	1.94%	3.36%	2.88%	52.14%	46.91%	57.40%
mean (22 benchmarks)		8.54%	9.01%	9.12%	2.42%	3.64%	2.32%	6.38%	4.25%	3.66%	45.27%	48.80%	53.00%

Miss rates: small-dm (L1D 64 sets x 1, L2 512 sets x 1)

		L1D			spill ld L1			spill st L1			L2		
benchmark	runs t/tr/r	test	train	ref	test	train	ref	test	train	ref	test	train	ref
perlbench	1/5/3	15.27%	10.44%	9.31%	0.00%	2.72%	2.07%	1.44%	5.74%	5.12%	0.34%	26.00%	20.19%
gcc	1/3/5	10.14%	13.26%	13.85%	2.27%	7.50%	7.23%	4.98%	8.51%	8.59%	32.44%	37.71%	47.89%
bwaves	2/2/4	44.12%	43.93%	44.55%	5.35%	6.13%	7.09%	1.37%	1.31%	1.13%	27.18%	26.52%	66.19%
mcf	1/1/1	28.22%	26.68%	24.46%	8.95%	10.49%	11.70%	10.17%	10.54%	9.98%	72.37%	71.66%	69.97%
cactuBSSN	1/1/1	17.36%	17.36%	16.94%	6.42%	6.42%	6.47%	5.51%	5.50%	5.01%	48.80%	48.81%	55.00%
namd	1/1/1	15.58%	15.53%	14.97%	12.11%	11.95%	6.99%	2.49%	2.50%	2.52%	37.33%	37.27%	32.32%
parest	1/1/1	15.36%	15.29%	15.19%	20.70%	13.83%	10.39%	6.63%	5.54%	5.80%	47.58%	50.52%	51.66%
povray	1/1/1	13.66%	14.30%	14.40%	13.93%	12.02%	9.21%	21.86%	21.55%	22.20%	37.82%	36.10%	37.67%
lbm	1/1/1	27.22%	26.83%	22.89%	1.63%	1.63%	1.68%	5.46%	5.34%	5.04%	75.06%	74.99%	78.54%
omnetpp	1/1/1	15.05%	18.54%	19.33%	0.72%	0.72%	12.85%	10.67%	10.69%	2.13%	36.64%	41.67%	37.69%
wrf	1/1/1	26.37%	25.82%	26.09%	13.69%	13.72%	13.66%	9.13%	8.46%	8.20%	26.50%	26.35%	29.03%
xalancbmk	1/1/1	15.16%	14.58%	18.92%	13.19%	30.23%	13.80%	18.14%	25.47%	17.84%	41.74%	26.71%	68.27%
x264	1/1/3	3.98%	4.04%	4.26%	0.98%	0.93%	0.83%	1.20%	1.20%	1.26%	26.05%	26.31%	25.90%
blender	1/1/1	9.92%	9.96%	10.33%	2.96%	6.67%	7.50%	7.02%	7.11%	3.36%	31.78%	27.63%	31.65%
cam4	1/1/1	14.02%	14.71%	15.12%	5.43%	5.03%	5.77%	4.41%	3.66%	4.62%	44.30%	46.92%	47.00%
deepsjeng	1/1/1	13.69%	13.57%	13.61%	6.73%	6.64%	6.44%	9.45%	9.69%	9.67%	19.74%	20.58%	14.36%
imagick	1/1/1	7.43%	5.62%	2.76%	16.95%	28.95%	6.34%	70.35%	2.48%	1.40%	47.92%	86.53%	68.78%
leela	1/1/1	8.96%	7.77%	7.98%	19.60%	12.09%	7.04%	0.46%	0.23%	0.28%	17.75%	15.93%	17.91%
nab	1/2/1	10.79%	11.36%	13.74%	1.54%	4.31%	3.93%	1.65%	3.90%	2.85%	15.30%	29.87%	31.34%
exchange2	1/1/1	4.01%	3.65%	3.11%	3.42%	2.78%	1.88%	3.67%	2.44%	2.01%	1.04%	1.48%	0.55%
fotonik3d	1/1/1	22.43%	22.47%	22.00%	9.89%	9.88%	9.79%	0.43%	0.43%	0.50%	38.96%	36.82%	49.28%
roms	1/0/1	30.01%	—	30.33%	7.11%	—	6.83%	3.38%	—	3.25%	38.90%	—	40.55%
xz	12/2/3	9.48%	12.39%	13.01%	2.97%	6.49%	4.61%	2.72%	6.05%	3.26%	28.59%	27.00%	35.37%
pooled		11.58%	14.84%	15.65%	3.60%	5.98%	5.43%	2.60%	4.80%	4.14%	35.05%	36.07%	45.16%
mean (22 benchmarks)		15.83%	15.82%	15.76%	7.70%	9.14%	7.15%	9.05%	6.74%	5.58%	34.33%	37.43%	41.66%

Miss rates: small-4way (L1D 16 sets x 4, L2 64 sets x 8)

		L1D			spill ld L1			spill st L1			L2		
benchmark	runs t/tr/r	test	train	ref	test	train	ref	test	train	ref	test	train	ref
perlbench	1/5/3	3.76%	5.22%	4.79%	0.00%	0.22%	0.09%	6.66%	3.20%	2.18%	0.00%	33.59%	21.67%
gcc	1/3/5	6.10%	8.58%	9.69%	0.33%	1.34%	1.69%	1.48%	4.09%	4.37%	36.94%	41.26%	53.56%
bwaves	2/2/4	40.95%	42.93%	43.60%	3.81%	4.92%	6.03%	0.02%	0.14%	0.06%	31.23%	26.80%	23.88%
mcf	1/1/1	24.85%	23.17%	21.53%	4.00%	5.26%	6.09%	7.21%	7.39%	6.44%	77.84%	77.15%	74.18%
cactuBSSN	1/1/1	15.30%	15.30%	15.63%	4.85%	4.85%	5.18%	4.27%	4.27%	4.30%	65.69%	65.68%	64.99%
namd	1/1/1	11.71%	11.59%	11.58%	2.78%	2.64%	2.40%	1.14%	1.23%	1.26%	28.28%	28.04%	27.50%
parest	1/1/1	10.21%	10.27%	10.38%	2.85%	2.03%	2.36%	1.61%	1.46%	1.67%	65.62%	69.50%	70.85%
povray	1/1/1	9.13%	9.02%	9.16%	6.99%	6.99%	7.21%	19.61%	19.46%	18.52%	45.58%	46.64%	45.94%
lbm	1/1/1	21.66%	21.30%	17.63%	0.00%	0.00%	0.01%	0.68%	0.50%	0.12%	97.36%	97.61%	97.01%
omnetpp	1/1/1	11.85%	15.94%	16.31%	0.05%	0.05%	0.08%	0.82%	0.83%	1.50%	28.01%	29.81%	28.08%
wrf	1/1/1	11.55%	11.05%	11.27%	6.88%	6.76%	6.63%	4.95%	4.70%	4.40%	34.19%	34.94%	33.77%
xalancbmk	1/1/1	13.28%	11.88%	17.12%	9.92%	15.33%	11.80%	18.32%	26.56%	22.71%	34.60%	12.81%	70.02%
x264	1/1/3	1.97%	1.97%	2.01%	0.22%	0.22%	0.17%	0.89%	0.88%	0.86%	31.00%	32.24%	32.92%
blender	1/1/1	5.06%	6.71%	8.00%	3.27%	5.33%	2.67%	3.87%	4.78%	1.42%	20.38%	12.80%	18.31%
cam4	1/1/1	9.07%	10.12%	10.69%	1.79%	1.77%	2.04%	2.70%	2.13%	2.35%	59.49%	59.86%	55.53%
deepsjeng	1/1/1	8.90%	8.71%	8.81%	2.01%	2.02%	2.10%	7.25%	7.46%	7.45%	7.54%	7.25%	6.35%
imagick	1/1/1	4.20%	5.01%	2.18%	8.47%	34.95%	3.93%	68.44%	0.48%	0.03%	85.57%	94.98%	96.60%
leela	1/1/1	5.85%	5.26%	4.94%	4.14%	1.74%	0.51%	0.42%	0.20%	0.00%	24.20%	20.07%	23.48%
nab	1/2/1	5.33%	4.97%	5.40%	0.51%	0.25%	0.13%	0.31%	0.34%	0.69%	6.02%	53.11%	59.18%
exchange2	1/1/1	2.01%	1.85%	1.67%	1.74%	1.40%	1.41%	2.98%	2.35%	2.24%	1.35%	1.76%	0.94%
fotonik3d	1/1/1	11.96%	11.67%	10.25%	3.82%	3.81%	4.03%	0.01%	0.01%	0.04%	61.79%	59.27%	66.95%
roms	1/0/1	14.94%	—	15.13%	3.05%	—	2.31%	0.79%	—	0.52%	53.16%	—	53.37%
xz	12/2/3	6.44%	8.32%	9.75%	0.96%	1.60%	2.08%	0.60%	1.21%	1.54%	26.86%	27.31%	34.20%
pooled		7.58%	10.27%	10.79%	1.85%	2.96%	3.07%	1.85%	3.12%	2.69%	44.14%	41.62%	47.07%
mean (22 benchmarks)		10.96%	11.40%	11.47%	3.15%	4.70%	3.12%	7.01%	4.26%	3.83%	39.52%	42.39%	45.72%

Miss rates: small-8way (L1D 8 sets x 8, L2 64 sets x 8)

		L1D			spill ld L1			spill st L1			L2		
benchmark	runs t/tr/r	test	train	ref	test	train	ref	test	train	ref	test	train	ref
perlbench	1/5/3	3.80%	4.87%	4.46%	0.00%	0.16%	0.09%	100.00%	3.50%	2.12%	0.00%	35.20%	22.93%
gcc	1/3/5	5.73%	8.09%	9.30%	0.23%	1.01%	1.21%	1.33%	3.93%	4.22%	38.80%	43.36%	55.50%
bwaves	2/2/4	26.02%	24.92%	22.29%	5.47%	4.82%	8.58%	0.02%	0.09%	0.04%	37.38%	36.16%	38.46%
mcf	1/1/1	24.64%	22.97%	21.44%	3.55%	4.49%	5.22%	6.92%	7.12%	6.19%	78.48%	77.82%	74.48%
cactuBSSN	1/1/1	15.62%	15.63%	15.86%	5.02%	5.02%	5.46%	4.67%	4.67%	4.62%	64.56%	64.55%	64.24%
namd	1/1/1	11.61%	11.48%	11.53%	2.55%	2.41%	2.43%	1.17%	1.25%	1.26%	28.50%	28.27%	27.58%
parest	1/1/1	9.90%	10.03%	9.94%	1.42%	1.20%	1.37%	1.47%	1.24%	1.33%	67.70%	71.14%	73.98%
povray	1/1/1	8.70%	8.58%	8.45%	6.93%	6.97%	7.25%	19.45%	19.27%	18.27%	47.73%	48.92%	49.55%
lbm	1/1/1	21.73%	21.37%	17.45%	0.00%	0.00%	0.00%	0.63%	0.46%	0.10%	97.05%	97.30%	97.87%
omnetpp	1/1/1	11.57%	15.90%	16.27%	0.05%	0.05%	0.08%	0.72%	0.73%	1.19%	28.59%	29.84%	28.12%
wrf	1/1/1	6.92%	6.62%	6.51%	6.23%	6.01%	5.81%	4.76%	4.45%	4.04%	55.78%	56.91%	57.32%
xalancbmk	1/1/1	13.12%	11.40%	16.84%	10.13%	12.13%	11.17%	18.90%	23.99%	20.10%	35.01%	13.19%	71.10%
x264	1/1/3	1.86%	1.85%	1.91%	0.16%	0.15%	0.12%	0.86%	0.85%	0.83%	32.44%	33.89%	34.26%
blender	1/1/1	5.04%	6.67%	7.97%	2.14%	4.89%	2.59%	3.87%	4.40%	1.31%	20.42%	12.85%	18.37%
cam4	1/1/1	8.84%	9.86%	10.43%	1.53%	1.52%	1.79%	2.52%	1.90%	2.13%	60.99%	61.38%	56.87%
deepsjeng	1/1/1	8.63%	8.41%	8.51%	1.77%	1.80%	1.85%	7.10%	7.31%	7.30%	7.70%	7.44%	6.50%
imagick	1/1/1	4.17%	5.01%	2.19%	7.19%	34.95%	3.59%	58.77%	0.40%	0.01%	86.22%	94.98%	100.00%
leela	1/1/1	5.16%	4.79%	4.20%	0.59%	0.25%	0.06%	0.42%	0.20%	0.00%	27.19%	21.89%	27.29%
nab	1/2/1	4.07%	4.04%	4.13%	0.16%	0.08%	0.09%	0.13%	0.26%	0.26%	7.87%	65.22%	77.32%
exchange2	1/1/1	1.34%	1.08%	0.96%	1.43%	1.02%	1.14%	2.70%	2.04%	1.80%	2.01%	2.98%	1.61%
fotonik3d	1/1/1	9.21%	8.73%	8.47%	2.50%	2.49%	2.73%	0.01%	0.01%	0.03%	80.25%	79.16%	81.02%
roms	1/0/1	11.83%	—	11.98%	2.45%	—	1.78%	0.89%	—	0.61%	67.05%	—	67.67%
xz	12/2/3	6.09%	7.86%	9.56%	0.42%	0.85%	1.26%	0.42%	0.86%	0.83%	28.35%	28.85%	34.84%
pooled		6.44%	8.99%	9.07%	1.80%	2.81%	3.08%	1.88%	3.21%	2.72%	49.51%	45.97%	54.37%
mean (22 benchmarks)		9.72%	10.01%	9.94%	2.70%	4.19%	2.90%	10.76%	4.04%	3.55%	42.41%	45.97%	49.96%

Formulas

column	formula	aciklama
spill/ROI	`((spill_load + spill_store) / roi_total_insts) * 100`	ROI komutlarinin yuzde kaci spill load veya spill store
spill ld/load	`(spill_load / roi_load) * 100`	ROI load'larinin yuzde kaci spill reload
spill st/store	`(spill_store / roi_store) * 100`	ROI store'larinin yuzde kaci spill store
(spill+csave)/ROI	`((spill_load + spill_store + csave_load + csave_store) / roi_total_insts) * 100`	ayni oran, callee-save save/restore dahil
(spill+csave) ld/load	`((spill_load + csave_load) / roi_load) * 100`	ROI load'larinin yuzde kaci spill reload veya callee-save restore
(spill+csave) st/store	`((spill_store + csave_store) / roi_store) * 100`	ROI store'larinin yuzde kaci spill store veya callee-save save
L1D	`(L1D misses / L1D accesses) * 100`	Tum load ve store access'lerinin (spill + other) L1D miss rate'i: L1D'nin bu konfigurasyondaki genel performansi.
spill ld L1	`(spill load L1D misses / spill load L1D accesses) * 100`	Spill load'larin yuzde kaci L1D'de miss oldu. Genelde dusuktur: spill slot'u kisa sure once yazildigi icin cache line cogu zaman hala L1D'dedir.
other ld L1	`((load misses - spill load misses) / (load accesses - spill load accesses)) * 100`	Other load'larin (spill olmayan load'lar) L1D miss rate'i. spill_ld_L1miss ile yan yana bakilir: spill load'lar normal veriden daha mi az, daha mi cok miss oluyor?
spill st L1	`(spill store L1D misses / spill store L1D accesses) * 100`	Spill store'larin yuzde kaci L1D'de miss oldu. Cache write-allocate: miss olan bir store once cache line'i L2'den getirir, sonra yazar.
other st L1	`((store misses - spill store misses) / (store accesses - spill store accesses)) * 100`	Other store'larin (spill olmayan store'lar) L1D miss rate'i; spill_st_L1miss ile karsilastirmak icin.
L2	`(L2 misses / L2 accesses) * 100`	L2'ye gelen tum isteklerin (L1D ve L1I miss'leri) yuzde kaci L2'de de miss oldu ve DRAM'e gitti.
spill ld L2	`(spill load L2 misses / spill load L2 accesses) * 100`	L1D'de miss olan spill load'larin yuzde kaci L2'de de miss oldu. Payda sadece L1D'yi miss eden spill load'lardir, butun spill load'lar degil.
spill st L2	`(spill store L2 misses / spill store L2 accesses) * 100`	L1D'de miss olan spill store'larin yuzde kaci L2'de de miss oldu (payda yine sadece L1D miss'leri).

spill = compiler-confirmed register-allocator spill (GCC `-dP`), csave = callee-saved register save/restore (CFI), both inside the ROI only. pooled = sum of numerators / sum of denominators over the completed runs; mean = average of the per-run percentages. Miss rates are per cache access; an L2 rate's denominator is the L1D misses of that access type.